LEARNER GUIDE

Formulae from words and diagrams

Formulae and Rearranging · Lesson 3. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Many everyday rules have the same shape: a **fixed amount** paid once, plus an amount for **each** mile, month or hour. A taxi costing £3 plus £2 a mile becomes C = 3 + 2m: the 3 is the fixed part, and 2m is 2 for each of the m miles. From a diagram, go round the shape once and add every side, then collect the like terms. Read descriptions in order: 'double x, then add 3' is $2x + 3$, but 'add 3 to x, then double' is $2(x + 3)$.

What you are learning to notice

- Find the fixed part (a number on its own) and the 'for each' part (a number × the letter).
- Choose a letter for each quantity, and say what it stands for: m for miles, n for months.
- The order of the words is the order of the operations. Brackets show what happens first.
- Test your formula with one value: does it give what the description says?

Moving through it, one step at a time

Underline the fixed part and circle the 'for each' part. Write the formula, then check it with a value such as 1 or 2 against the description. For a perimeter, write every side going round once, then collect the like terms. To use a formula, substitute in brackets and work in order.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — From words to a formula, then using it

A plumber charges £40 to call out, plus £25 an hour. Write a formula for the cost C of h hours, and find the cost of 2½ hours.

1. Fixed part: paid once.	40
2. 'For each' part: £25 for each hour.	25h
3. Put them together.	C = 40 + 25h
4. Substitute h = 2½: multiply first.	C = 40 + 25 × 2.5 = 40 + 62.5 = 102.5

C = 40 + 25h, and 2½ hours cost £102.50.

Example 2 — From a diagram to a formula

A rectangle is $(x + 2)$ cm long and x cm wide. Write a formula for its perimeter P.

1. Go round once, writing every side.	$P = (x + 2) + x + (x + 2) + x$
2. Collect the x terms.	$x + x + x + x = 4x$
3. Collect the numbers.	$2 + 2 = 4$

$P = 4x + 4$.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (writing simple expressions, and substituting values from Lesson 2), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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